

Putting the Count Back Into Accountability

Analyzing Transparency Data About the Sexual Exploitation of Minors

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Based on transparency disclosures for 17 US-based internet platforms and the national clearinghouse collecting legally mandated reports about child sexual exploitation (CSE), this study answers two research questions: What does the data tell us about the growth of online CSE? How reliable is that data? First, we analyze the growth in CSE reports over the last 25 years to derive a model and identify the underlying population. Second, we perform two comparative audits to assess the quality of transparency data. Critical findings include: First, the rapid growth of reports over the last decade is linear and largely driven by an equivalent growth in social media accounts. Second, the Covid-19 pandemic had no statistically relevant impact on yearly report volume. Third, while half of the surveyed organizations release meaningful transparency data, the other half either fail to make disclosures or release data with severe quality issues.

Additional Key Words and Phrases: social media, transparency reporting, audit, child sexual abuse materials, National Center for Missing and Exploited Children, CyberTipline

1 INTRODUCTION

Public debate about child sexual exploitation (CSE) features many adamant to alarming claims. There are the headlines in national newspapers announcing that “the internet is overrun with images of child sexual abuse” (Keller and Dance 2019), “Instagram connects vast pedophile network” (Horwitz and Blunt 2023), and “AI is about to make the online child sex abuse problem much worse” (Oremus 2024). There are Senate hearings on “big tech and the online child sexual exploitation crisis” featuring a who-is-who of social media CEOs (Durbin 2024) and laws referencing an “explosion in the multijurisdictional distribution of child pornography” (United States Congress 2008). There also are the Moms for Liberty, Qanon adherents, and Proud Boys mobilizing to “#SaveTheChildren” while

maligning everyone else, particularly LGBT folk, as “pedos” or “groomers” (Buntain et al. 2022; Martiny and Lawrence 2023). Even experts such as academics (Salter et al. 2023), lawyers (Haney 2021), hotline operators (O’Donnell 2021), or police (Europol 2020; Interpol 2020) are not above making lurid claims about the impact of the Covid-19 pandemic on CSE.

Given the gravity of the topic and the creeping sensationalism infusing even the law, two questions stand out:

- (1) What does available data tell us about the extent of online child sexual exploitation?
- (2) How accurate and trustworthy is that data?

Ready availability of accurate data can help ground public debate and experts’ contributions. It also is indispensable for designing or evaluating effective interventions, allocating content moderation and law enforcement resources, and predicting future trends and developments.

This article seeks to answer the two research questions based on transparency data about CyberTipline reports released by the technology firms filing the vast majority of them and the US not-for-profit receiving them, the National Center for Missing and Exploited Children (NCMEC). According to the US criminal code (United States Congress 2018, 2024), so-called electronic service providers (ESPs) must report all online CSE to NCMEC, lest they face financial penalties and lose their safe harbor protections against content liability. In turn, NCMEC triages reports, forwarding them to suitable law enforcement agencies, both within the United States and internationally. It also maintains databases with fingerprints of pictures and videos that qualify as child sexual abuse material (CSAM) and makes them available to service providers.

Each CyberTipline report describes an incident of child sexual exploitation, with the statutory “child” being any person under the age of 18, i.e., a *minor*. Reportable activities include the production, distribution, and possession of CSAM, sex trafficking of minors, sex tourism targeting minors, sexual molestation of minors, the online enticement of minors including grooming and sextortion, sending obscene material to minors, and misleading domain names as well as hosted content. Beyond capturing the statutory category, the schema for CyberTipline reports (National Center for Missing

and Exploited Children 2024) includes fields that describe the incident itself, e.g., date, time, platform, and involved persons, track the involvement of law enforcement, enable attachment of original content, and provide helpful context for triage, e.g., labels to identify memes, viral content, possible self-production, live streaming, generative AI, and the Tech Coalition classification (Tech Coalition 2022). The latter comprises four labels to distinguish between (A) pre-pubescent and (B) post-pubescent as well as (1) sex act and (2) lascivious exhibition. A few actual CyberTipline reports (without attachments) have become public as part of court proceedings (National Center for Missing and Exploited Children 2017a,b).

Three aspects of transparency disclosures about CyberTipline reports make them particularly well-suited to answering the two research questions. First, platforms' community guidelines differ substantially (Bateman et al. 2021; Jiang et al. 2020), are enforced inconsistently (Common 2020; Díaz and Hecht-Felella 2021), and are reflected in transparency disclosures to varying degrees (Urman and Makhortykh 2023). As a result, it is generally difficult, if not impossible, to aggregate or compare service providers' transparency data. However, since they are produced through a statutory mechanism, CyberTipline reports are an exception: Their counts *can* be aggregated over time and service providers; they also *can* be compared over time for the same provider or between (some) providers and NCMEC.

Second, content curation and moderation are key features of contemporary, centralized social media that engage in surveillance capitalism (Zuboff 2019) as a business model—if nothing else to keep the platform's content sufficiently inoffensive so that advertisers won't bail (Masnick 2022; Patel 2022). Alas, that same curation and moderation effort raises thorny questions about platform governance (Klonick 2018), with transparency reports serving as an important accountability mechanism (Access Now et al. 2021; European Parliament and Council 2022). Since social media also are the currently dominant means for dissemination of CSAM (Steel et al. 2020), their transparency reports are bound to capture much of that violative content as well.

Third, the legal requirement to file CyberTipline reports only applies to service providers operating out of the United States. At the same time, many American internet platforms have global reach. For

instance, at the end Of 2024, Meta alone had 3.9 billion daily active people across all its platforms—or 41% of the world population (Meta Platforms, Inc. 2025). Hence, their transparency data captures violative content and behaviors from all across the world. In fact, from 2019 to 2024, only between 5.2% and 15.8% out of the 17.0 to 36.2 million (M) reports received by NCMEC every year were about incidents in the United States; the vast majority happened outside the country.

At the same time, transparency data about CyberTipline reports is not a panacea when characterizing online CSE. It fundamentally depends on organizations noticing incidents of CSE, filing CyberTipline reports about them, and also making the attendant transparency disclosures. That is the case for most of the larger centralized platforms operated by US corporations. But it holds to a lesser degree for smaller platforms and federated social media, which lack resources and tools for comparable trust and safety efforts (Roth and Lai 2024). Finally, it does not hold for the dark web, which is designed for anonymity and hence for avoiding accountability. Still, when using appropriate methods, transparency data about CyberTipline reports can yield considerable insights.

2 METHODS

In discussing methods (here) and findings (below), we adhere to the two research questions for structuring the presentation. All data and Python

2.1 The Deluge of CyberTipline Reports

To characterize the extent of online CSE, we start by considering yearly CyberTipline report counts from inception of the tipline in 1998 to 2024 (inclusive, like all year ranges in this article) and then narrow temporal and geographical scopes for additional insight. In addition to utilizing familiar descriptive statistics, our analysis compares the goodness-of-fit for linear models, all but one using a single independent variable and report counts as dependent variable, based on adjusted R^2 . We first characterize overall growth in CyberTipline reports by comparing exponential growth, i.e., a linear model over the logarithm of the dependent variable, with piecewise-linear growth for the full year range. Next, to identify the likely underlying population driving that growth, we narrow the

temporal scope to 2014–2024 and compare linear models for human population, internet users, and social media accounts. To add detail, we narrow both temporal and geographical scopes and compare rankings of per-country report counts normalized by human population as well as social media accounts from 2020 to 2024. Finally, to explore the impact of the Covid-19 pandemic, we restore global scope and compare standard residuals for the linear growth model, using the customary two standard residuals as threshold for identifying outliers.

The primary source of data are NCMEC’s transparency disclosures to the congressional appropriations committees, which include total report counts going back to 1998 but otherwise cover only 2020–2024, as well its per-country breakdowns, which cover 2019–2024. Our analysis also draws on datasets about human population (United Nations), internet users (International Telecommunication Union), and social media accounts (DataReportal). Machine-readable files for all data and hyperlinks to their sources are included with the supplemental materials for this article (*citation elided for review*); we discuss data curation in more detail below.

2.2 Comparative Data Audits

To characterize the quality of transparency data, we validate that service providers’ transparency disclosures include meaningful metrics for online CSE and, if they do, assess to what degree repeated disclosures of the same quantities diverge in two different ways. The first such audit checks the *internal consistency* of computer-readable data, i.e., csv files, released by the same service provider. For each reporting period $p + 1$, it compares all entries shared with the previous period p , which are the entries for periods 1 to p making up the file for period p . Since this audit compares the same metrics published by the same organization some number of months apart, we expect that values only change as a result of deliberate modification, notably, to correct incomplete or buggy disclosures. Hence, differences should exhibit temporal or logical locality, i.e., be systemic errors. Descriptive statistics suffice for checking whether this is indeed the case.

The second audit checks the *external consistency* of CyberTipline report counts as disclosed by service providers and NCMEC in its per-provider breakdown. Since the latter provides only yearly

counts and, in 2019–2020, did so only for entire corporations, this audit also compares yearly per-corporation counts. Since this audit compares the results of two independent organizations tabulating the same unidirectional message flows, we expect some small degree of divergence that is proportional to the number of messages and caused by (non-deterministic) software bugs or hardware outages. Differences should not exhibit discernible patterns, i.e., be random errors.

We characterize concordance with the intra-class correlation coefficient (ICC) for count data (Carasco 2022). However, due to the small number of per-year, per-provider samples, with $1 \leq n \leq 6$ and $\bar{n} = 3.7$, we could only compute a goodness-of-fit for the entire dataset, across all nine providers, with $N = 33$. When comparing a Poisson model and negative binomial models with variance growing linearly as well as quadratically with mean, only the negative binomial with quadratic variance has an acceptable goodness-of-fit.

Since we cannot validate the model for individual providers, we also consider Bland-Altman or mean-difference plots (Bland and Altman 1995) for each provider individually and also all of them together. Consistent with the assumption that error for CyberTipline report counts grows with magnitude, we plot percentage differences instead of absolute values. We use the mean as denominator, since it provides an unbiased best guess for the true value in absence of other information. Furthermore, since the normality assumption does not hold for count data, we omit the customary 95% limits of agreement and instead trace the mean difference. We then use visual inspection to identify patterns and use the sign test to confirm their statistical significance.

We select candidates for the two audits based on three criteria, applied to service providers and their subsidiaries: Being a popular social media site based on Buffer’s yearly rankings from 2022–2024 (Lua 2022, 2023; Oladipo 2024), being labelled a very large online platform by the European Commission (European Commission 2023), and having filed 100,000 or more CyberTipline reports during any year, which are roughly 0.5% of the total report count for 2024. As listed in Table 1, 17 service providers meet at least one of the criteria. However, six of them do not make any transparency disclosures about online CSE. A further two service providers fall short of making semantically meaningful disclosures about CSE. One of the remaining nine service providers qualifies for the first audit

Table 1. An overview of surveyed service providers and their subsidiaries as well brands. Criteria for inclusion are being a popular social medium (Social), a very large platforms in the EU (XL), or a filer of $\geq 100,000$ reports in a year (100k). *Disclosures* identifies when providers started making disclosures about CSE and the current release frequency. *Audits* indicates how the validity of the provider's data was assessed.

Service Provider	Social	XL	100k	Disclosures		Audits
Amazon		✓		2020..	Y	Reports 2020–2024
Twitch	✓			H1 2020..	H	
Apple		✓				—
Automattic						—
Tumblr	✓					
Aylo (MindGeek)						Reports 2020–2024
Pornhub		✓		2020..	H	
Discord	✓		✓	H2 2020..	H	CSE Metrics
Google		✓	✓	H1 2020..	H	Reports 2020–2024
YouTube	✓	✓	✓	H1 2020..	H	
MediaLab						—
Imgur			✓			
Kik			✓			
Meta		✓	✓			Data Q3'21–Q4'24
Facebook	✓	✓	✓	Q3 2018..	Q	& Reports 2024
Instagram	✓	✓	✓	Q2 2019..	Q	
Threads	✓					
WhatsApp	✓		✓			
Microsoft		✓	✓	H1 2020..	H	Reports 2020–2023
LinkedIn	✓	✓		H1 2019..	H	
Skype	✓					
Teams	✓					
NCMEC				2019..	Y	Reports 2019–2024
Omegle			✓			—
Pinterest	✓	✓		H1 2020..	Q	Reports 2020–2023
Quora	✓					—
Reddit	✓		✓	2019..	H	Reports 2019–2024
Snap	✓	✓	✓	H2 2019..	H	Reports 2022–2023
TikTok	✓	✓	✓	Q1 2022..	Q	CSE Metrics
Wikimedia		✓				—
X (Twitter)	✓	✓	✓	H1 2024..	H	Reports 2024

comparing successively released csv files. All nine qualify for the second audit comparing CyberTipline report counts with NCMEC. Note that Omegle, a video chat service, shut down in November 2023 to settle a lawsuit by a minor who was sexually exploited on that platform (Loffhagen 2023).

The primary source of data are NCMEC’s yearly breakdowns of per-provider CyberTipline report counts as well as the nine service providers’ transparency reports. For detailed provenance of all datasets, we refer to the archive of supplemental materials for this article (*reference elided for review*). It includes all data utilized by this study in machine-readable form as well as all Python and R code for performing the analysis and illustrating results, all released under a permissive license. The archive also contains a supplement collecting hyperlinks for all transparency disclosures and datasets utilized by this study.

2.3 Data Curation

Perusal of even a few hyperlinks in that inventory makes clear that most of the disclosures are *not* machine-readable. To accurately capture the data in machine-readable form, we manually transcribed and validated disclosures with a small number of quantities. For larger tables, we relied on advanced editor features and ad-hoc scripts to restore tabular data extracted from HTML and PDF documents with copy & paste, while also making sure that the quantities themselves were preserved as is. For DataReportal’s social media accounts data, which is spread across approximately 1,200 slides formatted as SVG outlines, two people performed the manual data entry across three sessions on separate days. During a session, one person read each quantity out loud, and the other person entered the datum and read it back. For Meta’s csv files, which are not archived by the firm, we supplemented files downloaded from Meta’s website over the last three years with copies from the Internet Archive going back to Q2 2021.

In curating disclosures, we generally preserve a provider’s terminology and numerical representation. We make three exceptions that are grounded in CyberTipline reporting and hence have a well-defined meaning. They are *reports* for the number of CyberTipline reports that were filed by the provider, *pieces* for the number of pictures, videos, and other documents that were attached to CyberTipline reports, and *accounts* for the number of user accounts that were terminated for their involvement.

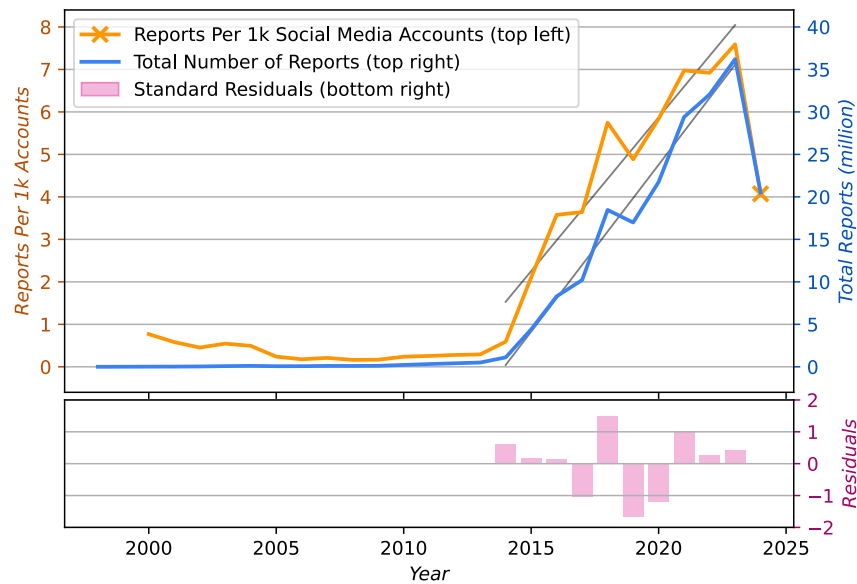


Fig. 1. Yearly CyberTipline reports (in blue, linear fit in gray, top-right y-axis), reports per 1,000 social media accounts (in orange, linear fit in gray, top-left y-axis), and the standard residuals from 2014–2023 (in magenta, bottom-right y-axis).

To map service providers’ metrics to the three standard categories, we make the following accommodations. First, we treat the common euphemism “permanent suspension” as account termination. Next, in lieu of a better match, we use CSAM counts as pieces. While that may fail to capture some violative content, 96.8% of all reports received by NCMEC in 2024 involve CSAM and at least 99.2% of all reports for the four year before do the same. Finally, for Meta, we treat its “child nudity & sexual exploitation” category as pieces. In Q2 2021, Meta switched to the more precise “child endangerment: sexual exploitation,” which we also treat as pieces. Including the earlier, imprecise category maximizes coverage for the service provider filing at least 91.5% of all reports in 2019–2021.

3 FINDINGS

3.1 The Deluge of CyberTipline Reports

Figure 1 shows three different views onto yearly CyberTipline report counts, each with a different scale. First, the blue line in the upper panel tracks the absolute number of reports in millions from 1998–2024, with the right y-axis covering the range from 0 to 40M reports. Second, the orange line

in the upper panel tracks the number of reports per 1,000 social media accounts, with the left y-axis covering the range from 0 to 8 reports per 1k accounts. For both curves, gray lines indicate the linear least squares fit for the decade from 2014 to 2023. Despite the different scales, both curves converge on the same point for 2024, hence the orange mark in addition to the blue line. Third, the magenta bars in the lower panel track the standard residuals for report counts during the same decade, with the right y-axis covering the range from -2 to 2 standard deviations. Each of the three views illustrates one of the following findings.

Finding 1. Yearly changes in the number of CyberTipline reports are best modelled as piecewise-linear with three distinct epochs, from 1998 to 2013, from 2014 to 2023, and for 2024.

A 2019 study by authors affiliated with Google, NCMEC, and Thorn characterized the growth of CyberTipline reports as exponential (Bursztein et al. 2019), and NCMEC has been doing the same in its transparency disclosures to the congressional appropriations committees for 2022, 2023, and 2024 (NCMEC 2023, 2024, 2025b). The assertion seems unwarranted after a 43.4% drop in report counts for 2024. Furthermore, even when excluding report counts for 2024, the fit of a piecewise-linear model with discontinuity in 2014 (adjusted $R^2 = 0.9911$) is better than that of an exponential model (adjusted $R^2 = 0.954$). Also, the mean squared error of the piecewise-linear model is 15× smaller than that of the exponential model.

NCMEC attributes much of the drop in 2024 to a change in reporting practices. Starting that year, service providers can file a single report for viral content and memes that qualify as CSAM instead of filing a report each time that same content is posted. That way, the 20.5M filed reports account for 29.2M incidents (NCMEC 2025a). NCMEC notes that the decline by 7M from the previous year's peak of 36.2M reports is unexpected, especially since child sex trafficking and online enticement became reportable that same year. It then blames Meta's deployment of end-to-end encryption (E2EE) for that difference. Alas, that underestimates the magnitude of the difference because if the growth of the prior decade had continued, 3.93 million more reports than in 2023 would have been filed in 2024, making for a difference of 10.9M reports.

Finding 2. The likely population driving the absolute yearly increase of 3.93 million reports over the decade from 2014 to 2023 is social media accounts. That does not, however, account for a relative increase from roughly one to almost eight reports per 1,000 social media user accounts over the same period.

When considering only 2014–2023, a linear model relating report counts to social media accounts as independent variable (adjusted $R^2 = 0.980$) has a better fit than that for years (adjusted $R^2 = 0.977$), world population (adjusted $R^2 = 0.966$), internet users (adjusted $R^2 = 0.964$) and years as well as social media accounts (adjusted $R^2 = 0.978$). Being reported for online CSE requires internet access, hence the model for social media accounts having a better fit than that for human population seems reasonable. Since the social media amongst surveyed platforms account for 92.6%–96.2% of all CyberTipline reports from 2019–2024, the model for social media accounts having a better fit than that for internet users also seems reasonable. However, we wouldn't necessarily expect the model for human population to have a better than that for internet users, even if the difference in adjusted R^2 is rather small. That raises the question of whether the differences in adjusted R^2 are meaningful.

To explore that question, we increase the resolution to per-country report counts, narrow the scope to 2020–2024, and compare reports per capita with reports per social media accounts. When ranking countries by the former, $9 \leq n \leq 14$, $\bar{n} = 12.2$ of the worst thirty countries are members of the Arab League, a phenomenon that has no obvious explanation and was prominently noted by Childlight (2024) as well. The fact that the country with the most reports per capita alternates between Libya and the United Arab Emirates (UAE) makes this even more puzzling. After all, the two countries are substantially different—with Libya spending much of the 2010s in a state of civil war (Global Conflict Tracker 2024) and the UAE being more stable and denser populated, while also having better infrastructure and larger GDP.

When considering reports per social media accounts, members of the Arab League continue to rank highly, with $10 \leq n \leq 14$, $\bar{n} = 12.8$ countries amongst the worst thirty from 2020 to 2024. At the same time, the distribution of rates does change, with yearly means of 5.2–8.7 reports per

1,000 social media accounts (r/ksoc) roughly *double* the yearly means of 2.8–4.6 reports per 1,000 capita (r/kcap). Meanwhile, the yearly maxima of 49.8–119.3 r/ksoc are roughly *triple* the maxima of 15.5–38.0 r/kcap. (We ignore a 13× larger rate for Tokelau in 2024 because the territory’s population, 2,506 people, is too small.) In other words, differences between countries with higher rates are more pronounced for social media accounts than human population.

In addition to being more clearly differentiated from lower ranked countries, the countries with the highest rates per social media accounts more consistently are troublespots. In 2020 and 2021, the country with the highest reports per social media accounts is Eritrea. While Afghanistan and Yemen are ranked second- and third-worst in 2021, they are the worst ranked two countries for 2022–2024. Common to all three countries is that they were recently engaged in (civil) wars ([Global Conflict Tracker 2025a,b,c](#)), with the attendant lawlessness apparently also making the online sexual exploitation of minors worse.

While worst-rated countries have rates in the double digits, rates for the vast majority of countries for both human population and social media accounts are consistently and noticeably smaller. Notably, at least 86.5% out of the 237 ranked countries have a rate of at most 4 r/kcap from 2020 to 2024. Likewise, at least 85.6% have a rate of at most 8 r/ksoc, which also equals the global rate for 2023.

Overall, social media filing the vast majority of CyberTipline reports weighs strongly in favor of treating social media accounts as the underlying population. Its per country ranking having a larger domain and more clearly differentiating countries with the highest rates is helpful when country rankings are required. That global and per country rates are closely aligned is reassuring. That the worst-ranked countries consistently are global troublespots, however, cuts both ways: The consistency can be seen as reassuring or as a flag for inaccurate data, since conducting representative population surveys in warzones would seem unusually fraught with threats to validity.

While social media accounts seems like a strong contender for population driving much of the growth in CyberTipline report counts, it definitely is not the only factor. After all, the normalized reports per social media accounts are not nearly constant and grow linearly from approximately 1 r/ksoc in 2014 to 8 r/ksoc in 2023. The drop down to 4 r/ksoc in 2024 suggests that a substantial

fraction of reports over the decade before that year is the result of memes and other unwelcome content that is not intended to sexually exploit children. While we lack the data for identifying other factors with confidence, we will note that the number of service providers filing reports has grown from 230 in 2021 to 294 in 2024. Moreover, service providers, including Meta (Meta 2023), Reddit (Reddit 2025), and Snap (Snap 2024), acknowledged improved detection of CSAM and more generally CSE in recent transparency disclosures, which points towards improved detection as a major remaining contributor to the increase.

Finding 3. The Covid-19 pandemic had no statistically relevant impact on the yearly volume of CyberTipline reports.

As illustrated by the magenta bars in Figure 1, the standard residuals for CyberTipline reports are not particularly large, ranging from -1.7 to 1.5 standard deviations. In other words, the data is noisy when compared to its model, but there are no pronounced outliers. Furthermore, with -1.2 and +1.0, the standard residuals for the pandemic years 2020 and 2021 are clearly smaller, i.e., well within the noise. Finally, when considering year-over-year growth, the increase of +0.5 standard deviations from 2019 to 2020 actually was on the smaller end. In other words, the growth in CyberTipline report volume for the pandemic years is consistent with the growth for the rest of the decade.

Even though no residual is larger than the customary two-standard-residuals threshold for outliers, the report count did increase by 2.221 standard residuals from 2020 to 2021. That is large enough to possibly include a substantial pandemic-induced increase: Under this scenario, content moderators had reduced reviewing capacity for much of 2020 and allowed a moderation backlog to build up. Towards the end of 2020, they finally found their pandemic routine, nimbly recovered from their slump, and started drawing down the backlog. As a direct result, they filed a good number of additional CyberTipline reports in 2021. Our analysis of the European Commission's Digital Services Act Transparency Database shows that a delay of roughly 100 days from content being posted to being actioned is average for protection of minors policies. Under the above scenario that delay would

grow about threefold, which does not seem unreasonable. Still, the number of additional CyberTipline reports that might have resulted from such a scenario is capped by the 2.221 standard residual difference between 2020 and 2021. As such, it statistically is within the noise.

We also reviewed previous work about the impact of the Covid-19 pandemic on the online sexual exploitation of minors. Besides a blog post from April 2021 by NCMEC that features only year-over-year comparisons (O'Donnell 2021) and a university's research report from May 2021 that nicely captures the pressures faced by individual hotline staff and police officers during the pandemic (Salter and Wong 2021), peer-reviewed articles (Dąbrowska 2021; Haney 2021; Salter et al. 2023) consistently cite reports by Europol from June 2020 (Europol 2020) and Interpol from September 2020 (Interpol 2020).

Both reports feature mostly anecdotal and indirect evidence. Both reports also misrepresent data about peer-to-peer networks by the Child Rescue Coalition. They claim that graphs of sinusoidal signals that happen to rise between February and April 2020 somehow demonstrate a “significant increase in the sharing of CSEAM” (child sexual exploitation and abuse material)—to use Interpol's boldface phrase. In addition to sharing the same sourcing and description, Interpol's single curve very much looks like the sum of Europol's four country-specific curves. That suggests that both police agencies are presenting the same data. However, according to y-axis labels, Interpol's counts are roughly 5,700× larger than Europol's aggregates, suggesting that the misrepresentation isn't the only problem with that data.

Having said that, Europol's report does contain compelling evidence that the pandemic did, in fact, result in noticeably more incidents involving CSAM, albeit *at a different timescale*. When looking at CyberTipline reports NCMEC forwarded to European law enforcement agencies during the first half of 2020 at *monthly* granularity, there is a 5× spike for March 2020 only. Given the date, the pandemic is the most plausible cause. Alas, the effect is not sustained enough to be detectable in yearly statistics.

Finding 4. The publicly available transparency data neither proves nor disproves that end-to-end encryption is a major cause for the steep decline of CyberTipline reports in 2024. But it does suggest a plausible alternative explanation.

In late 2023, just as Meta was starting to update its Messenger chat service to utilize E2EE by default, NCMEC published a blog post describing Meta’s decision to do so a “devastating blow to child protection” ([National Center for Missing and Exploited Children 2023](#)). Then, in its transparency disclosures for 2024, the organization bemoans the negative impact of E2EE on Meta’s report volume ([NCMEC 2025a,b](#)). The drop in Meta’s reports—from 30.7M reports in 2023 to 13.8M reports in 2024—certainly is steep. E2EE may very well have been a contributing factor. But the publicly available data points towards a more complex reality than NCMEC acknowledges.

To begin with, Meta’s WhatsApp service has been using end-to-end encryption since 2016; with 2 billion users compared to Messenger’s 1 billion, it also is the more popular chat service. Yet despite WhatsApp’s encryption, Meta has been moderating messages and reporting those involving CSE to NCMEC ([Elkind et al. 2021](#)). First, metadata is not encrypted and may suffice for detecting some suspicious activities, for example, when an adult keeps interacting with minors even though there are no familial or professional reasons for doing so. Second, when reporting another user, the WhatsApp client, which does have access to the unencrypted message content, forwards up to five messages sent by the user being reported to Meta’s content moderators. Messenger incorporates the same capability. The only difference is that Meta does not disclose the number of forwarded messages; for unencrypted chats, it is thirty.

NCMEC’s own transparency disclosures demonstrate that these monitoring mechanisms have impact in practice. The hotline started distinguishing between Meta’s brands in its transparency disclosures in 2021. Over the four years to 2024, WhatsApp submitted 1.4, 1.0, 1.4, and 1.9 million CyberTipline reports. These figures are substantial: Besides Meta, only Google, Snap, and TikTok filed more than a million reports in a year. They also represent a year-over-year increase of 33.2% for 2024. That implies that Facebook and Instagram are responsible for Meta’s overall decline that year.

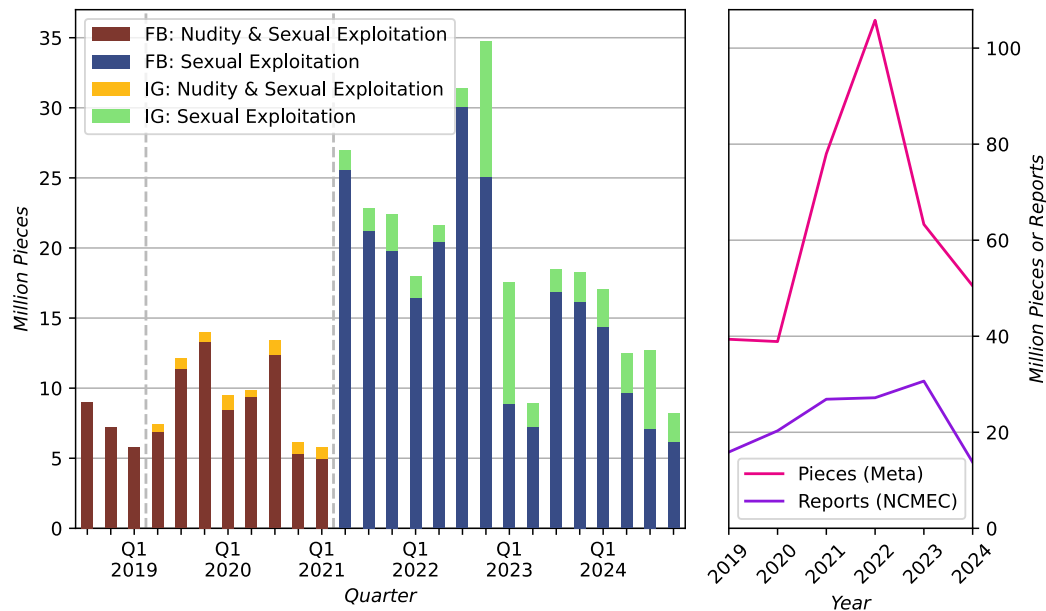


Fig. 2. Meta: Quarterly pieces (left) as well as yearly pieces and reports (right). The dashed lines indicates major changes to Meta’s metrics, with the firm starting to include Instagram’s pieces in Q2 2019 and switching to more precise metrics in Q2 2021 (see *Methods*).

Even if we are inclined to believe that the latter was mostly caused by E2EE, WhatsApp’s increase over the same time period happened *despite* E2EE. Furthermore, Snap deployed E2EE for photos and videos in 2019, yet has been filing more reports year-over-year for every year since (Teunissen and Napier 2022).

Meta’s disclosures of piece counts provide further evidence that E2EE may not be the primary cause of the overall decline in reports. As illustrated in the right panel of Figure 2, Meta’s overall piece count substantially declined in both 2023 as well as 2024. Meta started deploying E2EE in December 2023. So even if we exclude Q4 2023, the left panel of Figure 2 shows that pieces substantially declined year-over-year for the first three quarters, too—when Meta had not even begun deploying E2EE. The fact that reports increased while pieces decreased in 2023 implies a shift from image- and video-based CSE, notably CSAM, to other forms of CSE, notably online enticement. While E2EE does make detection of CSAM harder, metadata can help identify the latter by tracing inappropriate contact between adults and minors. In summary, none of this disproves the potential impact of E2EE. But

it also doesn't support E2EE having this impact and suggests some independent factor as plausible alternative.

3.2 Comparative Data Audits

3.2.1 *Meaningful Metrics.*

Finding 5. Discord's and TikTok's transparency disclosures suffer from poorly chosen metrics, which render them largely useless for tracking online CSE. They also use misleading language while introducing their respective metrics.

Discord does not disclose the number of CyberTipline reports filed but the number of accounts reported to NCMEC. That quantity is 34%–83% smaller than the number of reports. While that is an interesting statistic, its disclosure without additional context is problematic. First, since fewer reported accounts than reports implies that the same accounts were reported repeatedly, one wonders how they were able to do so more than once given the firm's oft emphasized "zero tolerance" for CSE. Likewise, one wonders why the metric was picked. Its introduction in the transparency disclosure for H2 2020 only reinforces that concern. The quantity was introduced in a section titled "Reports to NCMEC" with a bar chart titled "H2 2020 NCMEC Reports." Only the paragraph below the chart clarified that Discord reported "6,948 *accounts* to NCMEC" (emphasis ours).

TikTok discloses absolute counts for the number of actioned videos only; all other metrics such as the breakdown into categories and subcategories of violative content are expressed as decimal fractions. Before introduction of a new schema in Q2 2023, the firm did not disclose all subcategory fractions, omitting the fraction of videos actioned by automation, and thereby made it impossible to calculate the number of videos actioned because of CSAM. Upon introduction of the new schema, TikTok claimed that it "added granularity." While it did increase the number of subcategories from 26 to 29, it also removed subcategories, notably reducing minor safety subcategories from five to four. That change means that it remains impossible to calculate the number of videos actioned because of

CSAM. Meanwhile, TikTok’s csv files diverge in capitalization for labels and serialization of floating point numbers, e.g., the same quantity may appear as “0.9620000000000001” and “0.962.”

3.2.2 *Internal Consistency.*

Finding 6. In addition to being the only service provider to disclose only rounded quantities, an average of 3.6% of Meta’s transparency disclosures from Q3 2021 to Q1 2023 diverges quarter-over-quarter. Divergent entries vary widely in metrics and quarters, with no apparent temporal or logical locality.

For seven quarters, from Q3 2021 to Q1 2023, Meta’s csv file for that quarter diverges from the previous quarter’s csv file for $0.6\% \leq n \leq 7.3\%$, $\bar{n} = 3.6\%$ of all entries. Divergent entries vary widely in policy area and actual metric (e.g., content actioned, appealed, or restored). Notably, 12, 3, 7, 7, 11, 17, and 4 entries about CSE differ from the previous quarter over the seven quarters, respectively. Divergent entries also vary widely in the affected period, going back up to 7–11 quarters every quarter but Q4 2021 and Q1 2023. The latter two quarters also are the quarters with the fewest divergent entries, i.e., 1.2% and 0.6%, respectively. Overall, there is no discernible temporal or logical locality to these errors. Meta never acknowledged these discrepancies. Though they did stop just as the EU’s Digital Services Act was taking effect.

Alas, the actual extent of divergent entries across Meta’s transparency data is unknowable. Since Meta is the only service provider to round its transparency data, additional disparities may very well have been erased by the rounding process. By the same argument, we cannot exclude that Meta’s data continues to suffer from quality issues. They may just be small enough not to survive rounding. The presence of disparities also raises the question of why Meta decided to round transparency statistics in the first place.

3.2.3 *External Consistency.* Figure 3 shows the mean difference plots for the nine service providers that release CyberTipline report counts, first individually for each provider across nine panels and then for all providers in the tenth panel. For each pair of report counts disclosed by a service provider

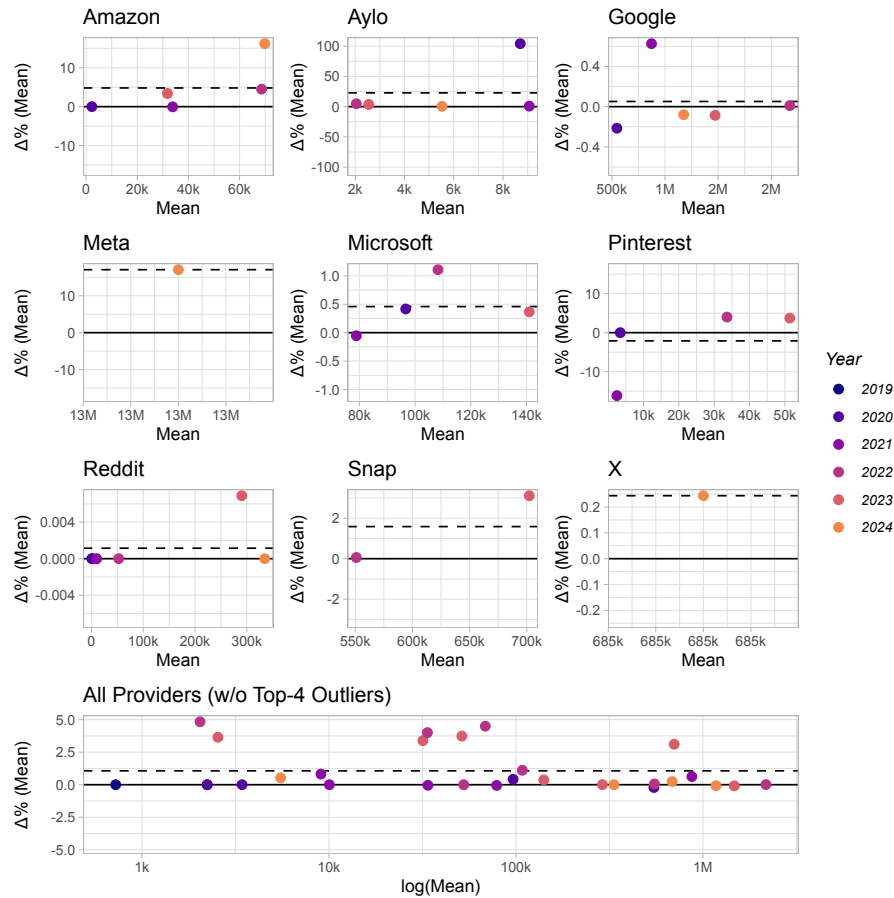


Fig. 3. Mean difference plots for nine service providers individually and all of them together. Each plot has its own scales, and the x-axis is logarithmic for the tenth plot. For readability, the tenth plot omit's the top-4 outliers, i.e., one point each for Amazon, Aylo, Meta, and Pinterest.

and NCMEC, the mean difference plots include a point with x the mean of the two counts, y the difference of the two counts as a percentage of the mean, and the color reflecting the year. If the service provider's count is smaller than NCMEC's, the percent difference is positive (and vice versa). The dashed line is the mean of the percentage differences. Each of the ten plots has its own dimensions for axes and the tenth plot's x-axis is logarithmic. For readability, the tenth plot omit's the top-4 outliers, i.e., one point each for Amazon, Aylo, Meta, and Pinterest.

Finding 7. Aylo's Pornhub is the only service provider to acknowledge and explain a mistake in their handling of CyberTipline reports.

When considering y-axis scales, the plot for one provider sticks out: Aylo's 104.1% difference for 2020. The firm only makes transparency disclosures for its most visible brand, Pornhub. In its transparency report for the year, which was published in April 2021, the brand not only acknowledges the discrepancy but also explains it as the result of submitting the same reports "multiple times in an abundance of caution." They add that NCMEC has been notified of the duplicate reports. Strikingly, this is the only instance we could find of a service provider acknowledging a problem with their disclosures or providing a credible explanation for the problem.

Finding 8. For the nine service providers that disclose at least a year of CyberTipline report counts, all but 4 out of 33 counts are within 5% of NCMEC's disclosures. They also exhibit a very small but statistically relevant bias towards underreporting.

In addition to Aylo's outlier, there are three considerably smaller ones at 16.2% for Amazon, 17.1% for Meta, and -16.1% for Pinterest. We have no additional insight into Amazon's and Pinterest's outliers. But we do have an informed guess for Meta's outlier: The firm does not usually include WhatsApp in its transparency disclosures. Sure enough, if we add WhatsApp's report count for 2024 as disclosed by NCMEC to the report count disclosed by Meta, the percent difference ceases to be an outlier, coming down to 2.3%. That would also bring it into range with the remaining 29 percent differences, all of which have magnitude smaller than 5%. That's small enough to conclude that the other eight service providers—Amazon, Aylo, Google, Microsoft, Pinterest, Reddit, and Snap—make a good faith effort at data collection and disclosure. Then again, even if we include the outliers, the intra-class correlation coefficient is 0.9999, indicating very strong concordance.

Meanwhile, the mean difference plots make another pattern clearly discernible, namely that the majority of percentage differences is positive. That holds for individual values, with 20 positive, 7 zero, and 6 negative percentage differences. That also holds for the means of the percentage differences, which are indicated by a dashed line in each panel and positive for eight out of nine providers as well as for all providers together. Even when disregarding the four outliers, the remaining 29

percent differences still have a positive mean of 1.1%. The sign test further validates that this bias is statistically significant with a p-value of 0.0047.

In theory, positive percentage differences are either the result of providers undercounting or NCMEC overcounting or both. In practice, it seems unlikely that NCMEC would overcount for all providers but Reddit, which has the same yearly report counts as NCMEC across five out of six years and five out of seven equal pairs. In either case, with 1.1% on average ignoring the four outliers, the respective bias is very small.

4 DISCUSSION

In trying to make approximate sense of the above findings, we start by noting that NCMEC's very mission is the protection of minors, whereas content moderation is a means to an end for service providers, i.e., to ensure that their platforms remain attractive to advertisers. Anything beyond that, including CyberTipline reporting or transparency disclosures, probably requires external incentives to justify the spend. We consider three scenarios for service providers, each with its own incentives, and then turn to NCMEC. In doing so, we identify larger take-away lessons.

The first scenario is based on the observation that US law does not mandate affirmative scanning for CSAM and other forms of CSE. However, once a service provider "obtains actual knowledge of any facts or circumstances" involving CSE, they must file a CyberTipline report "as soon as reasonably possible." (§2258A of [United States Congress \(2024\)](#)). The same statutory section also imposes strict record keeping and confidentiality requirements on service providers. Give the general abhorrence of CSE, the administrative overhead of CyberTipline reporting may incentivize some providers to try ignoring CSE on their platforms. The problem with such a laissez-faire approach to content moderation is that, in this day and age, it seems pretty much impossible to maintain plausible deniability about CSAM and CSE. That way, laissez-faire turns into willful negligence ([Salter and Sokolov 2024](#)), which can pierce safe harbor protections in case of sex trafficking. For instance, the lawsuit that caused Omegle to shut down alleges negligence in half of its claims ([Long et al. 2022](#)).

4.1 Predicting the Future

The second scenario is based on the very real threat of losing safe harbor protections in addition to financial penalties and a shattered reputation. When service providers recognize that the combined threat is existential, content moderation and CyberTipline reporting become the obviously preferable options. Furthermore, once these controls are in place, the marginal cost for service providers filing an extra CyberTipline report approaches zero, especially with increased automation. That cost differential becomes a powerful incentive to report any and all content and actions that could remotely be construed as CSE, including memes or content shared in outrage (Grossman et al. 2024). In other words, service providers are strongly incentivized to overreport. That, in turn, poses a major challenge for NCMEC, which needs to process the combined report streams of all service providers.

Take-Away 1. With almost all reports encompassing CSAM, it is too coarse a criterion for categorizing the content of CyberTipline reports. More granular accounting is urgently needed.

Now, NCMEC enjoyed a reprieve from the crushing absolute number of CyberTipline reports in 2024, down 43.5% from the peak of 36.2M reports the previous year. The question is how long will that reprieve last? We answer the question based on two different ways of relating the old and new reporting regimes. The first way is the obvious one, i.e., it is based on the ratio between the compressed number of reports, i.e., 20.5M and the uncompressed number of incidents, i.e., 29.2M. That gives us a scaling factor $\sigma_1 = 20.5/29.2 = 0.702$. Alas, we already know that this scaling factor is on the strict side, as it accounts for memes and similar content only.

The second way utilizes NCMEC's coarse ranking of reports as either low-quality *informational* or high-quality *actionable* content, which happens to be embedded in a table that also categorizes reports by type of law enforcement recipient, for example, see NCMEC (2025b). We can leverage this ranking because reports that capture more than one incident also are informational. According to NCMEC's disclosures, the percentage fractions of actionable reports in 2020–2024 are 46.1%, 62.6%, 49.1%, 50.3%, and 78.4%, respectively. Just as expected, the percentage fraction of actionable reports increases substantially in 2024. More interestingly, in the years 2020, 2022, and 2023, the percentage

fractions of actionable reports are close, with $\bar{n} = 48.5\%$. We ignore the outlier in 2021 and treat the mean as the real percentage fraction for the old reporting regime. To derive the second scaling-factor, we translate the fraction of informational reports for the new regime to the equivalent fraction for the old regime (the informational component) and add it to the fraction for actionable reports for the same regime (the actionable component). That yields $\sigma_2 = 48.5\% + 48.5\%/78.4\% * 21.6\% = 0.618$.

With the two scaling factors in place, we can now adjust the yearly report rate $r = 3.93\text{M}$ from our model for 2014–2023 to project growth from 2025 onward to be between $\sigma_2 r = 2.43\text{M}$ and $\sigma_1 r = 2.76\text{M}$ reports per year. If we assume that 40M reports is NCMEC’s breaking point, that gives the organization between 7 and 8 years until CyberTipline reports come close to overwhelming its capacity again. This does assume that social media accounts keep growing and CSE detection keeps improving at current rates. If either slows down, then we need to adjust the model and projections as well.

Take-Away 2. As illustrated by these projections, meaningful analysis requires incorporating events such as decline over 40% of CyberTipline reports for 2024. But doing so requires additional data, such as the relationship between reports and incidents, and not just for the one abnormal year.

4.2 Obstructing the Past

The third scenario considers what to do *after* all CSE has been detected and all CyberTipline reports have been filed. Large service providers with a substantial presence in the European Union are subject to that country block’s Digital Services Act (DSA) (European Parliament and Council 2022), which mandates transparency disclosures of aggregate statistics on an at least yearly basis and of individual decisions, suitably stripped of personally identifying information, on a daily basis.

We might expect that this substantial, but localized legal requirement would encourage US service providers to rationalize their operation by instituting the same transparency practices globally. So far, however, the Brussels effect does not seem to have taken hold for transparency disclosures (Bradford 2020). Besides the fact that the DSA only took effect in 2023, two factors seriously complicate the release of transparency disclosures. First, the audience for transparency data is generally ill-defined.

That means that service providers have little guidance on what metrics to disclose. The Santa Clara Principles ([Access Now et al. 2021](#)) and DSA provide some guidance but are surprisingly high-level and imprecise.

Second, while the audience for transparency disclosures remains relatively ill-defined, the negative reputational impact of transparency disclosures can be stark. We already highlighted some of the more sensationalist claims about CSE in the introduction to this article. They result, in part, from statistically unencumbered cherry-picking of transparency data, notably, mistaking absolute CyberTipline report counts as a measure of prevalence, and an information ecosystem that favors emotionally charged, viral, and conspiratorial content. The combination thereof serves as a strong incentive for service providers to withhold data or cherry-pick metrics themselves. While there is some measure of grim satisfaction in watching social media being hoisted by their own petard, the stakes for CSE are simply too high and the consequences too grave to leave it at that.

In particular, the ill-defined requirements for transparency disclosures also complicate validation thereof. Meta provides a telling example: The firm commissioned two external audits. First, a panel of academic experts performed a qualitative review of Meta’s policies, enforcement process, and transparency reporting and made several on-point recommendations for their improvement ([Bradford et al. 2019](#)). Second, Ernst & Young performed a quantitative assessment of Meta’s transparency data collection ([Meta 2022](#)). According to Meta, they went through “in-depth preparation” themselves and provided E&Y with full access to data as well as data scientists, software engineers, and internal auditors. Yet both Meta and E&Y didn’t think to compare metrics with those from the previous quarter and thereby missed $27/2,192 = 1.2\%$ divergent entries for Q4 2021.

Meta also nicely illustrates the outcomes from this tension between transparency and self-protection. The firm probably has invested more than any other service provider into trust and safety and makes a number of regular transparency disclosures. But it also has quite the history of releasing faulty or misleading metrics and favoring profits over minor safety. To list some highlights, Meta repeatedly misreported advertising metrics to customers ([Bruell and Patel 2020](#); [Vranica and Marshall 2016](#)),

platform usage data to a researchers (Timberg 2021), and advertises in its transparency database (Rosenberg 2019). When the latter resulted in a string of critical press reports, Meta deplatformed the academics who had collected the data (Edelson and McCoy 2021). It also shut down its CrowdTangle transparency tool when it became inconvenient (Kerr 2024). As far as minor safety is concerned, the firm ran an ad campaign sexualizing teenage girls and targeting middle-aged straight men (for TikTok nonetheless) (Silverman and Mac 2020), tried to expand Instagram's teenage audience despite knowing that the platform is toxic to teenage girls (Wells et al. 2021), and generally was reluctant to even consider minor safety (Sung and Silberling 2024).

Meanwhile, this study also illustrates the potential of transparency data. It critically relies on CyberTipline reporting imposing a uniform process on participants and on NCMEC having markedly different incentives. Perhaps reassuringly, the findings demonstrate that a sizable number of service providers, i.e., half of surveyed organizations, act in good faith when disclosing transparency data.

Take-Away 3. Suitably designed regulation is uniquely suitable for establishing a shared vocabulary for transparency disclosures and hence enables broader analysis. When combined with redundant disclosures, ideally involving several independent parties, it also facilitates validation of disclosures.

The usual discourse about service providers and CSE seems to be based on an assumption that, given enough content moderation, online CSE can be minimized if not eradicated. This article is certainly based on the assumption that, given enough data on the spread of CSE, we can understand the spread—and then develop more effective strategies for containing it. This narrative has certainly been embraced by social media platforms. They often treat transparency disclosures as opportunities for touting the effectiveness of their stewardship of our online interactions.

The findings, however, suggest that the relationship between social media and CSE is more complicated, we dare say, mutually reinforcing to some degree. If we accept that US-based, corporate social media are designed for intense emotional engagement and viral content, why would we expect that they have a different impact on people with a pathological interest in children? If we accept that social media have made it easier to show pictures and videos to a potentially global audience than

any other technology, why would we expect that they have a different impact on CSAM? The rational answer is, of course, that we shouldn't expect much of a difference.

Consider the growth in CyberTipline reports. Our analysis suggests that social media accounts are the likely underlying population driving that growth from 2014 onwards. But it doesn't explain why this period of hypergrowth started in that particular year. Alas, observers of internet culture do: They identify just that year as the year social media consolidated their hold on us (Cao 2024; Staltz 2017). A similarly strange phenomenon is that Arab League countries consistently have high rates of reports per capita and social media accounts. Given the differences between member countries, it stands to reason that some cultural factor is behind these elevated rates. We leave identification of this factor for future work, but will note that it may very well be extrinsic, e.g., caused by prejudice of content moderators.

Take-Away 4. Reigning in the spread of online CSE without structural changes to social media may just be impossible.

4.3 Exaggerating the Present

Besides operating the CyberTipline, NCMEC also helps locate missing children (its original mission), maintains databases of CSAM hashes as well as victims, and offers support services to victims, their families, and law enforcement. It was chartered by law, is largely funded by the US government, and works closely with law enforcement. For these reasons, courts have found NCMEC to be a "government agent" that must abide by Fourth Amendment protections against unreasonable search (Grossman et al. 2024), which complicates operation of the CyberTipline. At the same time, NCMEC is not a law enforcement agency. Instead it probably is best understood as an organization that seeks to protect a uniquely vulnerable population, children, from some of the worst crimes and, failing that, to support the victims and their families in their recovery.

That goes a long way towards explaining why NCMEC uses too broad a brush at times when describing the impact of CSE. Examples from this study include the insistence on characterizing the

growth in CyberTipline reports as exponential even after a substantial decline, or the reductive declaration that Meta's drop in reported incidents must be caused by E2EE. At the same time, the only reason that this study could complicate the picture for the examples are NCMEC's transparency disclosures. In fact, since 2022, when we started this study, NCMEC has been improving its transparency disclosures every year by enlarging the scope with additional metrics and by adding granularity to existing statistics. That is an excellent track record, especially when compared to service providers, and we sure hope NCMEC continues doing so.

Take-Away 5. Given its globally significant role, NCMEC's transparency disclosures are uniquely important for understanding the spread of online CSE. Their continued release and improvement are a major contribution.

5 CONCLUSION

In summary, the evidentiary record for CyberTipline reports does *not* support the more sensationalist claims about online child sexual exploitation. There is *no* "explosion" of CSAM. The internet is *not* "overrun with images of child sexual abuse." When taking yearly growth into account, the pandemic did *not* trigger more sexual exploitation overall, even if there was a marked spike in March 2020.

Nonetheless, the evidentiary record does support the claim that US-based social media, with their unprecedented global reach and propensity for virality, are the biggest driving force behind the growth of CyberTipline reports over the decade from 2014 to 2023. While that growth threatened to overwhelm NCMEC, i.e., the very organization serving as a critical global resource, improvements to the reporting of repeated incidents combined with an unexpected decline in reports for 2024 have resulted in a reprieve. Our projections based on a linear model for growth and two different metrics for the decline in 2024 suggest that NCMEC will be at breaking point in 7 to 8 years.

As far as transparency is concerned, many technology firms did better than we expected. As this study has amply demonstrated, transparency disclosures needn't be mere transparency theater, but can help produce real insight. The most surprising result from this study is the finding that firms

generally disclose reasonably accurate CyberTipline report counts but nonetheless have a very small but statistically significant bias towards underreporting. It would appear that the professionalism of the trust and safety personnel preparing these statistics prevails, but cannot entirely overcome all too human biases. That seems like a very human result on a topic where most counts represent devastating human implications.

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